Rates & Ratios Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Flour is sold in 4 different-sized packs. Which pack is the best buy? 1

A. 250 g pack for $0.70

B. 500 g pack for $1.30

C. 1 kg pack for $2.30

D. 2 kg pack for $4.80

Convert each option to 250g or equivalent

Option A: 250g for $0.70

Option B: 500g for $1.30 = 250g for $0.65 (÷2, ÷2)

Option C: 1kg for $2.30 = 250g for $0.575 (÷4, ÷4)

Option D: 2kg for $4.80 = 250g for $0.60 (÷8, ÷8)

∴ option C is the best buy

2 Eliza had saved $300 to use on fuel for a road trip during her holidays. Her car has a fuel consumption of 6.4 $L/100\ km$ and she paid $1.98 per litre for fuel throughout the trip. How many kilometres was Eliza able to travel using the money she'd saved? Answer correct to the nearest kilometre. 2

Litres of fuel she can afford = $300 ÷ $1.98

= 151.515... L

6.4L for 100 km

÷6.4 ÷6.4

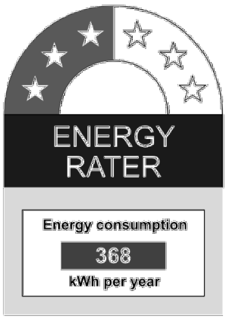
1L for 15.625 km

×151.515.. ×151.515...

151.515... 2367.42...

She can travel 2367 km (nearest km)

3 A 2500-watt vacuum cleaner has the following energy label on it. The energy label assumes that a typical household would consume 368 kWh of energy per year using the vacuum cleaner. How many hours and minutes of usage would this equate to each week? Give your answer correct to the nearest minute. Assume 52 weeks per year. 3



Assumed number of kWh per week = $\frac{368}{52}$

= 7.076... kWh

energy rating of vacuum = 2500 W = 2.5 kW

Power consumption = power rating × time

7.076... = 2.5 × t

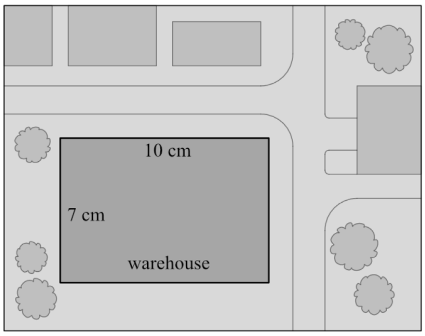
t = $\frac{7.076...}{2.5}$

= 2.8307... h

= 2 h 49 min 50.77 sec.

= 2 hours 50 mins (nearest min)

4 A company buys a new warehouse to improve their manufacturing capacity. They plan to save money by collecting rainwater from the entire area of the rectangular roof of the warehouse. The diagram below shows a plan view of the warehouse and its location within its surroundings. The dimensions of the warehouse on the diagram are 10 cm and 7 cm.



a. If 10 cm on the diagram represents 100 m, what would be the scale of the diagram? 1

Express the scale as a ratio in simplest form.

10cm = 100m
10cm = 10000 cm
∴ scale = 10 : 10000
= 1 : 1000

b. If the company's goal is to collect 1200 kL of water from the roof, how much rain must fall? Answer in millimetres, correct to the nearest millimetre. 3

Actual roof measurements : 70m and 100m
area of roof = 70 × 100 = 7000 m²
V = 1200 kL = 1200 m³
V = Ah
1200 = 7000 × h
$\frac{1200}{7000} = h$
h = 0.171428... m
= 171 mm (nearest mm)

5 The ratio of fiction to non-fiction books in a library is 4:7. If there are 484 fiction books, how many books are non-fiction? 1

$$\frac{484}{4} \times 7 = 847$$

6 The fuel consumption for a car is 7.6 liters/100 km. On a road trip, the car travels a distance of 1860 km. If the cost of fuel is \$1.85/litre, calculate the total fuel cost for this trip? 2

$$1860 \div 100 \times 7.6 \times 1.85 = 261.52$$

7 The table shows the average energy used in kilojoules per kilogram of body mass, when cycling at two different speeds for 30 minutes. Shane weighs 62 kg. If 1 calorie is equivalent to 4.184 kilojoules, approximately how many more calories would Shane use, cycling at 20 km/h, than he would at 15 km/h during the 30 minutes? Answer correct to the nearest whole number. 2

Cycle Speed	Average energy used (kJ/kg) in 30 minutes
15 km/h	11.52
20 km/h	18.43

$$(18.43 - 11.52) \times 62 \div 4.184 = 102.3948 \ldots$$
$$\approx 102$$

8 120 fish were captured and all of them were tagged then released back into the ocean. One year later, 80 fish were captured of which 6 were tagged. Using the capture-recapture method, estimate the popluation of fish in the ocean. 2

Let x be the total population of fish in the ocean

Find the fraction of fish that was tagged after each time

First Capture: $\frac{120}{x}$

120 out of the entire population were tagged

Second Capture: $\frac{6}{80}$

6 out of the 80 that was caught were tagged

$$\frac{120}{x} = \frac{6}{80}$$

$$80 \times 120 = 6 \times x$$

$$9600 = 6 \times x$$

$$\therefore x = 9600 \div 6$$

$$\boxed{x = 1600 \text{ fish}}$$

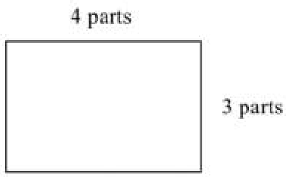
9 A rectangle has a perimeter of 56 cm. If the ratio of its length to its width is 4:3, find the area of the rectangle. 2

Total perimeter is 14 parts = 56 cm

1 part = 4 cm

So the dimensions are 16 cm by 12 cm.

The area = $16 \times 12 = 192\text{ cm}^2$



10 The scale of a map is 1 cm = 125 m. If the actual distance between two points is 650 m, what is the distance between these points on the map? 1

$\frac{650}{125} = 5.2$

11 The amount of a particular medication in an intravenous (IV) solution is 250 mg per 15 mL. A patient is prescribed 3500 mg of the medication. How much IV solution should be given to the patient? 1

$\frac{3500}{250} = 14$

$14 \times 15 = 210$ mL

12 Dominic's car uses fuel at a rate of 7.2 L per 100 km for long-distance driving and 10.4 L per 100 km for short-distance driving. Dominic drove his car 851 km to Melbourne. The journey included 112 km of short-distance driving. How much fuel did Dominic's car use on the journey, correct to the nearest litre? 2

112 Short: 1.12×10.4

739 long: 7.39×7.2

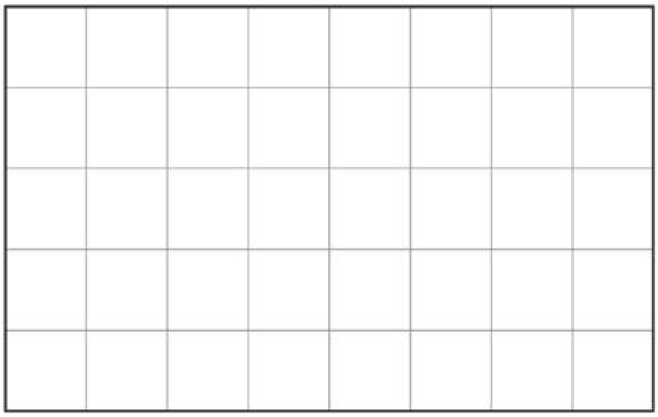
65 L

13 Richard plays on his game console for three hours every day. The game console uses 350 watts of energy per hour. The cost of electricity usage is \$0.45 per kWh. Calculate the yearly cost for Richard to play his game console, correct to the nearest cent. 2

$0.35 \times 3 \times 365 \times 0.45 = \$172.46$

14 A rectangular sportsground has been drawn to scale on a 1- cm grid as shown. The scale used is 1 : 3000 4

Kerry took 12 minutes to walk around the perimeter of this sportsground. What was Kerry's average speed in kilometres per hour?



$5 \times 3000 = 15000$ cm in real life

$8 \times 3000 = 24\,000$ cm in real life

Total distance $= 2\times(15000 + 24000)$

$= 78000$ cm $= 0{\cdot}78$ km

12 minutes is $0{\cdot}2$ hours

speed $= 0{\cdot}78 \div 0{\cdot}2 = 3{\cdot}9$ km/h

15 The sizes of the angles of a triangle are in the ratio 2 : 3 : 5. Find the size of the largest angle. 1

Total parts $= 2+3+5 = 10$

$180 \div 10 = 18°$ each part

largest angle $= 18 \times 5 = 90°$

Rates & Ratios Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 What speed is equivalent to 70 km/h? 1

A. 1.2 m/s B. 19.4 m/s

C. 252 m/s D. 1167 m/s

$$\frac{70 \times 1000}{60^2}$$

(B) 19·4 m/s

2 Filip drives a car with a petrol consumption of 8.3 L/100 km. The gas tank of the car has a capacity of 40 L. If Filip is driving 870 km, how many full tanks of petrol will he need to reach his destination? 2

$$\text{Petrol used} = \frac{870}{100} \times 8.3$$
$$= 72.21 \text{ L}$$

So Filip will need to use 2 full tanks of petrol

3 In the region of a tropical forest, 96 rare birds were captured, tagged and released. Five months later, 72 birds were recaptured from the same region and it was found that 18 of these birds had previously been tagged. Use the 'capture-recapture' method to give an estimate of the number of rare birds in this region of the forest. 3

First : 96 birds Tagged

2nd : 72 birds recaptured

18 tagged

$$\frac{18}{72} = \frac{1}{4}$$

$\therefore 96 \times 4 = 384$

4 What is the scale 1 cm to 5m equivalent to? 1

(C) 1 : 500

5 Maya was exercising, checked her pulse and counted 32 beats in 10 seconds. How could Maya's pulse be calculated in beats/minute? 1

A. $32 \div 10 \div 60$ B. $32 \times 10 \div 60$

C. $32 \div 10 \times 60$ D. $32 \times 10 \times 60$

(C)

6 For lunch, Mark has a chicken and avocado wrap with an energy content of 1972 kJ. When he rides his bike, he burns 1850 kJ energy per hour. How many minutes must he ride his bike to work off the energy from the chicken and avocado wrap. 2

1972/1850 x 60 = 64 mins

7 Sam has a household fridge that uses energy at a rate of 738 kWh per year. She is thinking of buying a bar fridge to store extra bottles of wine. The cost of the bar fridge is $420 and it has an energy consumption rate of 313 kWh per year.

a. If energy is charged at the rate of $0.32/kWh, what is the energy cost of running the bar fridge for one year? $0\cdot32(313) = \$100\cdot16$ 2

b. What would be the percentage increase in energy costs to run both fridges compared to running only the household fridge? Give your answer correct to the nearest percent. 2

$0\cdot32(738) = \$236\cdot16$

Both fridges $= \$336\cdot32$

$\dfrac{336\cdot32 - 236\cdot16}{236\cdot16} \times 100\% \doteq 42\%$

c. What would be the total cost of buying the bar fridge and running it for five years? 2

$420 + 5(100\cdot16) = \$920\cdot80$

8 Tim owns a Holden Commodore that runs on ULP and has a fuel consumption of 8·9L/100 km. When converted to run on liquid petroleum gas (LPG), the fuel consumption is 12·3L/100 km. Tim averages driving 18 000 km per year.

a. Calculate the annual cost for each type of fuel (assuming that the car only runs on one type of fuel for a year) if the average price of ULP is 195.9 c/L and LPG is 101.6 c/L. 2

ULP

$8\cdot9 \times \dfrac{18\,000}{100} \times 195\cdot9 = \$3138\cdot32$

LPG

$12\cdot3 \times \dfrac{18\,000}{100} \times 101\cdot6 = \$2249\cdot42$

b. How much does Tim save in fuel costs for the year by converting his car to LPG? 1

$3138\cdot32 - 2249\cdot42 = \$888\cdot90$

c. How much does he save per month? $\dfrac{888\cdot90}{12} = \$74\cdot08$ 1

d. The cost of converting the car to LPG is $2600. How many months would Tim take to break even? 1

$\dfrac{2600}{74\cdot08} = 35\cdot097\ldots$

$\therefore$ It would take 36 months

e. What distance would Tim travel before reaching the break-even point? 1

$\dfrac{18\,000}{12} \times 36 = 54\,000\text{km}$

9 The table shows the amount of water used in a variety of situations. Water costs \$1.95/kilolitre. Find the cost of water used if Thea showers for 6 minutes, does two loads of washing clothes, in a washing machine, washes his dog with a hose and washes three cars with a bucket. 3

Water usage	Volume	Time
Washing a dog with a hose	120 L	per dog
Shower	160 L	per 10 mins
Washing a car with a hose	350 L	per car
Washing a car with a bucket	80 L	per car
Washing clothes in a washing machine	150 L	per load

Shower: 160L for 10 mins
i.e 16L for 1 min. For 6 mins i.e. 16 x 6 = 96L
Washing clothes: 150 x 2 = 300L
Washing the dog: 120L
Washing the cars: 80 x 3 = 240L
Total Litres used = 756 L
Cost: 756/1000 x \$1.95 = \$1.47

10 a. Lance has a solar panel system installed in his house. How much power will this generate in 365 days if the average daily production of the solar panel system is 7.25 kWh? 1

7.25 x 365 = 2 646.25 kWh

b. Lance's average annual household consumption is 9856 kWh. What percentage of the household consumption is generated by the solar panel system? Answer correct to one decimal place. 1

% generated by solar panel = 2 646.25/9 856 x 100% = 26.8%

c. Lance also wants to know how much it costs to charge his laptop each year. He uses a 120 W charger for his laptop and it takes about 30 minutes. He does this five times a week, every week of the year. If the cost of electricity is \$0.456725/kWh, calculate the cost of charging his laptop for six months of the year. 2

Laptop power used for 6 months =120 W x 30/60 x 5 x 26 = 7800 Wh = 7.8kWh

Total cost for 6 months = 7.8 x 0.456725 = \$3.56

12 The current world record for the men's 100-metre sprint is 9.58 seconds set by Usain Bolt, a Jamaican sprinter, in 2009. What is the rate of Bolt's record-setting sprint in kilometres per hour, correct to one decimal place? 1

$$\begin{aligned} 100\text{ m} &= \frac{100}{1000} \\ &= 0.1\text{ km} \\ 1\text{ hour} &= 60 \times 60 \\ &= 3600\text{ seconds} \end{aligned}$$

$$\frac{0.1}{9.58} \times 3600 = 37.5782\ldots \approx 37.6\text{ km/h}$$